OKC Thunder Analyst Intern Project

Project Overview

The goal of this project was to build a model that predicts the probability that an NBA shot will be made based on information available about the shot, shooter, defenders, location, and game situation.

The final model produces a probability between 0 and 1 for each shot in the testing dataset. These predictions are provided in submission.csv using the required shot_id and make_prob columns.

The training dataset contains 425,719 shots, while the testing dataset contains 213,977 shots.

Data Exploration

The training data contains information about the shot itself, the offensive and defensive teams, the shooter and defenders, shot location, game state, shot clock, player movement, and defensive tracking.

Some of the most relevant variables include:

- Shot type
- Shot distance
- Shot location (locationx, locationy)
- Whether the shot was contested
- Number of contesting defenders
- Closest defender distance
- Closest defender approach
- Defender distances at different time points
- Shooter speed
- Shot clock
- Number of dribbles before the shot
- Game state
- Shooter and defender IDs
- Offensive and defensive team IDs

The training data contained 195,009 made shots and 230,710 missed shots.

I also examined categorical variables such as shot type, game state, and whether the shot was contested, along with numerical summaries of the shot and player-tracking variables.

Feature Engineering

In addition to the original variables, I created additional features intended to capture relationships that may be useful when predicting shot outcomes.

One important area was defensive tracking.

The dataset contains information about the closest defender's movement and distance. I converted the defensive tracking information into features representing defender distance at different time points before the shot.

I also created additional distance-based features, including squared shot distance, to allow the model to capture nonlinear relationships between distance and shooting probability.

These features were designed to provide the model with more information about both the shot itself and the defensive environment surrounding the shot.

Baseline Model

I first established a baseline model so that later improvements could be measured against a simple starting point.

The baseline model achieved a validation log-loss of:

0.65618

Log-loss was used because the objective is to predict probabilities rather than simply classify shots as makes or misses. A prediction of 0.80 means something different from a prediction of 0.55, so evaluating the quality of the predicted probabilities is important.

Lower log-loss indicates better probabilistic predictions.

Model Development

I progressively added additional features and model complexity to evaluate whether they improved predictive performance.

An intermediate model using additional categorical and engineered features achieved:

Log-loss: 0.64317

I then incorporated defensive-tracking features, which further improved the validation result to approximately:

Log-loss: 0.63997

This indicated that information about the defender and defensive movement provided additional predictive information beyond the basic shot characteristics.

Final Model

For the final model, I used CatBoost, a gradient-boosting model that can effectively handle a mixture of numerical and categorical variables.

The final model was trained using the engineered shot, player, location, game-state, and defensive-tracking features.

The model achieved a validation log-loss of:

0.62084

The best validation performance occurred at iteration 399.

This represented a substantial improvement over the original baseline log-loss of 0.65618.

After selecting the final model configuration, I retrained the model using the full training dataset and generated predictions for all 213,977 shots in the testing dataset.

Feature Importance

The final model's feature importance provided insight into which variables contributed most to its predictions.

The highest-ranked features included:

Feature	Importance
Shot type	15.34
Distance	14.75
Closest defender distance	10.02
Distance squared	9.34
Location X	5.83
Shooter speed	5.63
Shooter ID	4.92
Number of contesters	4.62

Shot clock	3.65
Defender distance at 0.25 seconds	3.04

Shot type and shot distance were the two most important features. Defensive proximity was also highly important, with closest defender distance ranking third.

This suggests that the model is using both offensive shot characteristics and the defensive environment when estimating make probability.

Test-Set Predictions

The final model generated a predicted make probability for each of the 213,977 test shots.

The resulting submission.csv contains two columns:

- shot_id
- make_prob

For example, each row represents one test shot and the model's estimated probability that the shot was made.

These probabilities can potentially be used to compare expected and actual performance, evaluate shot difficulty, and identify situations where the predicted probability differs substantially from the eventual outcome.

Interpretation and Potential Basketball Applications

A shot probability model can provide more information than simply identifying whether a shot was made.

For example, a shot with a predicted make probability of 0.70 can be considered a fundamentally different outcome from a shot with a predicted probability of 0.25, even though both ultimately result in either a make or a miss.

A model like this could potentially support basketball analysis by estimating expected shooting outcomes while accounting for factors such as distance, shot type, location, shooter movement, and defensive pressure.

The defensive-tracking features are particularly useful because they provide information about the difficulty of the shot beyond basic distance and location.

Limitations

There are several limitations to this approach.

First, the model is limited to the variables available in the provided dataset. Additional information about player context, defensive schemes, play type, lineup combinations, fatigue, and other game circumstances could potentially improve predictions.

Second, feature importance does not necessarily imply causation. A feature being highly important to the model means that it was useful for prediction, not necessarily that changing that feature would directly cause the probability of a make to change by a specific amount.

Third, player and team identifiers may allow the model to learn historical differences between players and teams. This can improve predictive performance but may also make it more difficult to generalize to situations involving players or teams with limited historical data.

Conclusion

The final model improved substantially over the baseline by combining shot characteristics, player and team information, spatial features, and defensive-tracking information.

The validation log-loss improved from **0.65618** for the baseline model to **0.62084** for the final CatBoost model.

The resulting model was then trained on the complete training dataset and used to generate make probabilities for all 213,977 test shots.

The project demonstrated the process of taking a large sports dataset from initial exploration through feature engineering, probabilistic modeling, model evaluation, interpretation, and final prediction generation.